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Longitudinal Changes in Hepcidin and Erythroferrone in Critical Illness

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Anemia After Critical Illness

- Anemia is a common complication after major surgery and critical illness.
- Anemia of critical illness is often accompanied by profound inflammation, resulting in alterations in iron availability and utilization that delay hemoglobin recovery.
- Practical Anemia Bundle for Sustained Blood Recovery (Warner et al, JAMA Netw Open, 2025)
 - Question: Does a multifaceted anemia prevention and treatment bundle vs. standard care improve hemoglobin recovery and functional outcomes in 100 adults with postsurgical or medical critical illness?

Original Investigation | Critical Care Medicine

Practical Anemia Bundle and Hemoglobin Recovery in Critical Illness A Randomized Clinical Trial

JAMA
Network | **Open.**

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The Intervention Bundle



Optimized Phlebotomy Practices

Low-volume blood draws, bundled laboratory timings, and closed-loop blood sampling



Clinical Decision Support

EHR alerts, daily rounding checklist – *minimize phlebotomy burden, ensure nutrition and appropriate bleeding prophylaxis, remove invasive lines*



Pharmacological Anemia Treatment

1,000 mg IV iron



The Bundle Improved Hemoglobin Recovery

PRIMARY OUTCOME

+0.7 g/dL

Adjusted mean difference in hemoglobin at 1 month (95% CI 0.13–1.20)

12.2 g/dL vs. 11.5 g/dL

Intervention

Standard Care

SECONDARY FINDINGS

- Fewer post-discharge transfusions
2.0% vs. 11.8%
- Fewer hospital readmissions
24.5% vs. 31.4%
- Favorable trend across fatigue, quality of life, physical function and cognition



Longitudinal Markers

Longitudinal changes in haemoglobin, iron stores, and inflammatory markers following surgery and in critical illness: an analysis from the Practical Anaemia Bundle for Sustained Blood Recovery randomised clinical trial
BJA

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- Iron assays, erythropoietin levels, hemoglobin, and inflammatory markers were obtained at enrollment and 1 and 3 months post-hospitalization.

1) Ferritin and TSAT correlated with inflammation (IL6, CRP), in opposite directions

2) Inflammatory markers positively correlated with erythropoietin and negatively correlated with hemoglobin

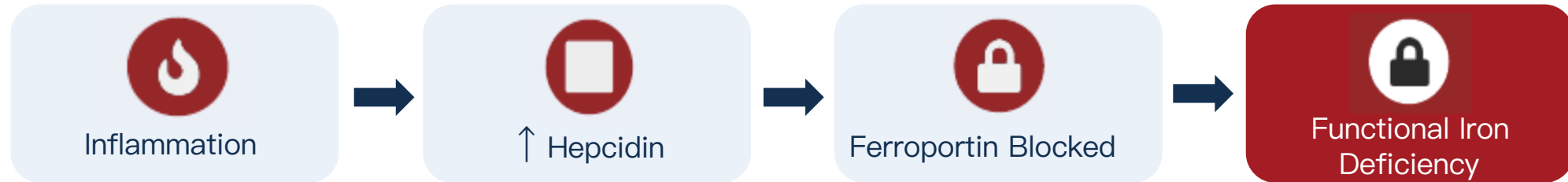
3) The soluble transferrin receptor–log ferritin index identified patients likely to experience the greatest hemoglobin responses to IV iron.

– sTfR–FI values and 1 –month Hb recovery amongst those receiving IV iron

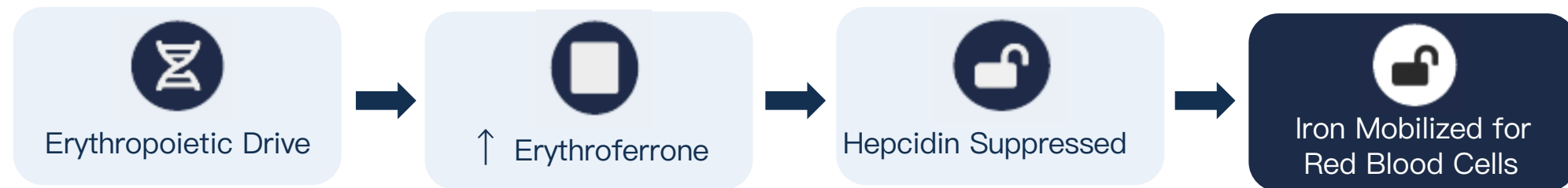
		Estimate (95% CI)	
sTfR–FI	<1	0.36 (-0.5, 1.21)	p=0.412
	1.5 +	2.03 (0.64, 3.43)	p=0.004

Hepcidin and Erythroferrone

Hepcidin
The Iron Lock



Erythroferrone
The Iron Key



Objectives & Methods

Primary Objective

Explore temporal changes of hepcidin and erythroferrone in this randomized clinical trial and assess correlations between these markers and traditional iron assays, erythropoietin levels, and inflammatory markers.

Secondary Objective

Explore IV iron-mediated hemoglobin responses by baseline hepcidin and ERFE levels.

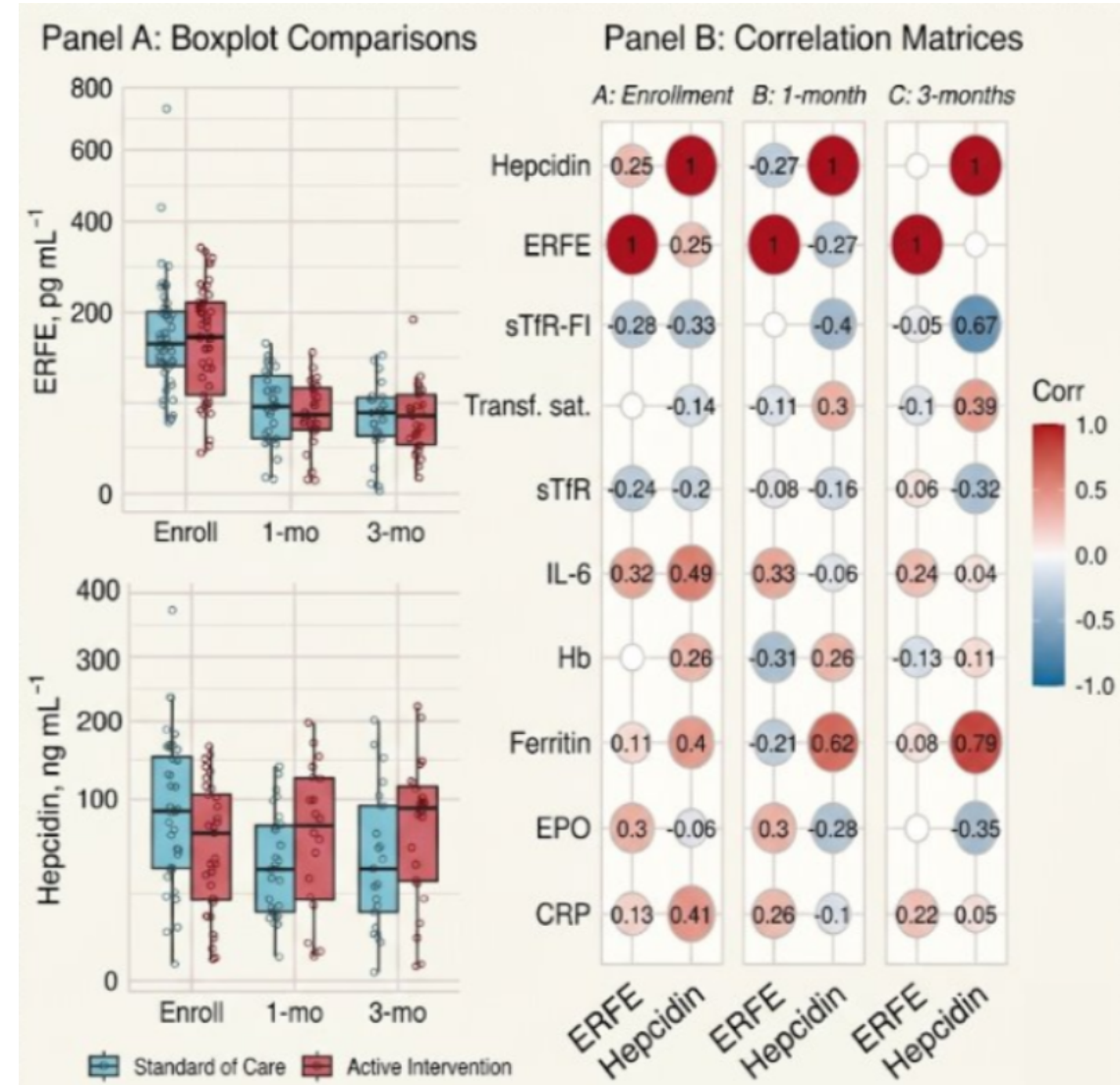
Methods

Hepcidin and ERFE levels — in addition to traditional iron assays, EPO levels, hemoglobin, and inflammatory markers — were obtained at enrollment (during ICU admission), 1 month, and 3 months post-hospitalization.



Findings

- Both hepcidin and ERFE were upregulated at enrollment during critical illness.
 - Concurrent activation of inflammation-mediated iron restriction and erythropoietic-drive
 - Note: 60% of patients experienced major bleeding prior to enrollment (≥ 500 ml blood loss + 2 g/dL Hb drop) $\rightarrow$ ERFE elevation; counteracted by inflammation-induced hepcidin elevation
- ERFE levels decreased over time, while no significant differences were observed for hepcidin (remained elevated at 3-months post-critical illness)
- Hepcidin and ERFE levels positively correlated with various inflammatory markers.
- Directionality of correlation for hepcidin and ERFE reversed at 1 month (positive $\rightarrow$ negative)



Findings cont.

Lab measure	Among low values	Among high values	Differential effect
ERFE	1.06 (-0.03, 2.15)	0.30 (-0.57, 1.17)	-0.76 (-1.82, 0.30)
Hepcidin	1.63 (0.44, 2.81)	1.97 (0.84, 3.09)	0.34 (-0.88, 1.56)

- Unlike the soluble transferrin receptor–log ferritin index, hepcidin and ERFE levels at enrollment did not identify patients likely to experience greater hemoglobin responses to IV iron.

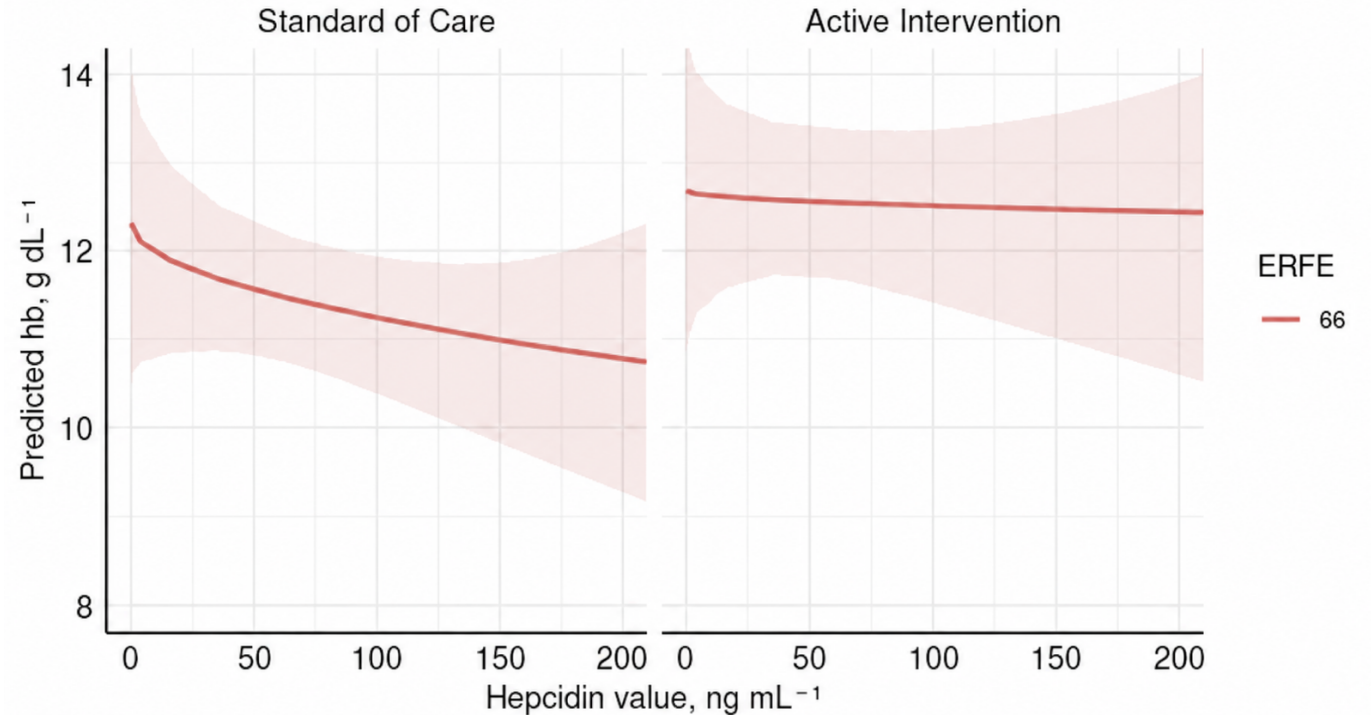


Findings cont.

- Amongst those with low ERFE levels, higher hepcidin levels were associated with lower 1 – month Hb responses
- This was attenuated in those randomized to IV iron
 - IV iron overcomes hepcidin elevation

Three-Way Interaction: Hepcidin & ERFE by Treatment

Predicted 1-m Hb for 65 y/o female with baseline Hb = 8.8



Conclusion

1

Hepcidin and ERFE are both elevated early in critical illness.

2

Suggests concurrent activation of inflammation-mediated iron restriction and erythropoietic drive.

3

Hepcidin and ERFE did not predict hemoglobin response to IV therapy; though IV iron may overcome hepcidin blockade in those with low baseline ERFE values

